

SOLAR POWERED IoT BASED ROBOT FOR RAILWAY TRACK INVESTIGATION WITH AI ASSISTANCE

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY



ANNA UNIVERSITY::CHENNAI 600 025

MARCH 2026

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TRACK INVESTIGATION WITH AI ASSISTANCE**

UEE1811 PROJECT WORK

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Under the Guidance of

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BONA FIDE CERTIFICATE

Certified that this project report titled **SOLAR-POWERED IoT BASED ROBOT FOR RAILWAY TRACK INVESTIGATION WITH AI ASSISTANCE** is the bona fide work of **GOPIKA S (8115U22EE014), SARIKA SRI K (8115U22EE041), SURIYADHARSHINI S P (8115U22EE049), VENNILA SRI S (8115U22EE055)**, who carried out the work under my supervision.

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DECLARATION BY THE PROJECT BATCH

We declare that to the best of my knowledge the project work reported here in has been composed solely by our project batch and that it has not been in whole or in part in any previous application for the award of a degree.

Submitted for the Project Viva Voce held at K. Ramakrishnan College of Engineering on _____

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INSTITUTE VISION AND MISSION

VISION:

“To achieve a prominent position among the top technical institutions”

MISSION:

- To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting-edge technologies.
- To provide education for developing high-quality professionals to transform the society.

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VISION:

To emerge as a renowned department for high quality teaching, learning and research in the domain of Electrical and Electronics Engineering producing professional engineers, to meet the challenges of society.

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PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

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PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES (COs):

COURSE OUTCOMES	KL	DESCRIPTION
C412.1	K1 & K2	On Completion of the project work students will be in a position to take up any challenging practical problems and find solution by formulating proper methodology.
C412.2	K1 & K3	Students are able to apply the fundamental knowledge of Electrical and Electronics Engineering in developing novel products/solutions and thereby contributing to society
C412.3	K1 & K3	Students become capable of designing and developing system prototypes independently by utilizing latest software's and equipments
C412.4	K1 & K2	Intellectual capability and innovative thinking of the students are ignited
C412.5	K3 & K5	By team work students are able to develop professionalism, build self-confidence and practice ethical responsibilities

COURSE OUTCOMES VS PROGRAM OUTCOMES MAPPING

(DETAILED; HIGH: 3; MEDIUM: 2; LOW: 1):

COURSE OUTCOMES	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
C412.1	3.00	3.00	0.00	1.00	0.00	0.00	1.00	2.00	3.00	0.00	0.00	0.00	1.00	2.00
C412.2	3.00	3.00	2.00	3.00	0.00	3.00	1.00	0.00	3.00	3.00	0.00	2.00	2.00	1.00
C412.3	3.00	0.00	3.00	3.00	3.00	0.00	0.00	3.00	3.00	3.00	1.00	2.00	1.00	1.00
C412.4	3.00	0.00	0.00	0.00	0.00	0.00	3.00	3.00	3.00	3.00	2.00	3.00	1.00	2.00
C412.5	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.00	3.00	0.00	3.00	1.00	1.00
C412	3.00	3.00	2.50	2.33	3.00	3.00	1.67	2.67	3.00	3.00	1.50	2.50	1.20	1.40

* For Entire Course, PO /PSO Mapping; 1 (Low); 2(Medium); 3(High) Contribution to PO/PSO

Type of Project: Product development

Domain: Renewable Energy /Control & Instrumentation/ IoT Applications

Real-World Factors Considered:

Factor	Justification
Environment	The system uses solar power, reducing dependence on non-renewable energy sources and minimizing environmental impact. It is designed to operate in outdoor railway environments with varying weather conditions.
Safety	The project enhances railway safety by detecting obstacles in advance, helping prevent accidents involving humans, animals, or objects on the track.
Ethics	The system is designed to protect human life and reduce harm without causing privacy violations, as it focuses only on obstacle detection on railway tracks.
Cost	12000
Standards Followed	IEEE standards for IoT communication and embedded systems, and general safety guidelines inspired by railway standards (Bureau of Indian Standards – BIS).

PO Mapping with level (Low/Medium/High)

PO No	Description	Level
PO1	Engineering Knowledge	High
PO2	Problem Analysis	High
PO3	Design & Development	High
PO4	Investigation	Medium
PO5	Modern Tool Usage	High
PO6	Engineer & Society	Medium
PO7	Environment & Sustainability	High
PO8	Ethics	Medium
PO9	Team Work	High
PO10	Communication	Medium
PO11	Project Management	Medium
PO12	Life-long Learning	Medium

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We thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

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ABSTRACT

Railway safety is very important because many accidents happen due to obstacles present on the track and late detection. To overcome this problem, we have designed a solar-powered IoT-based railway safety robot which works as a pre-checking system before the train reaches a particular location. In our project, the robot is controlled remotely using a mobile phone or laptop with the help of IoT, so it is not fully automatic but can be easily operated by a user. The main purpose of this robot is to move about 2 to 3 km ahead of the train on the same track and check whether any obstacle is present. For detecting obstacles, we use two sensors, IR sensor and ultrasonic sensor, which together form a sensor fusion system and help in detecting objects more accurately. Along with this, we use ESP32-CAM modules which are placed on all four sides of the robot to capture images and videos from different directions. Whenever an obstacle is detected, the camera captures the live view and sends it to a display screen placed in front of the co-pilot in the train control panel, so that the co-pilot can clearly see the track condition directly. A GPS module is also used to track the location of the robot and identify the exact position of the obstacle, which helps in better monitoring. In addition to this, we use an ESP8266 Node MCU as the main controller, which handles communication, data processing, and IoT connectivity. The entire system is powered using solar energy, where the energy from the solar panel is stored in a battery and used to run the robot continuously. Overall, this project helps in improving railway safety by detecting obstacles early, providing real-time monitoring, and reducing the chances of accidents in an efficient and simple way.

RELEVANCE TO SUSTAINABLE DEVELOPMENT GOALS (SDGs)

PROJECT TITLE	SDGs ADDRESSED
SOLAR-POWERED IoT BASED ROBOT FOR RAILWAY TRACK INVESTIGATION WITH AI ASSISTANCE	SDG 9: Industry, Innovation and Infrastructure
	SDG 11: Sustainable Cities and Communities
	SDG 7: Affordable and Clean Energy

SDG 9: Industry, Innovation and Infrastructure

RELEVANCE: This project uses modern technologies like IoT, AI, and robotics to improve railway infrastructure and safety systems.

IMPACT: It helps in protecting human lives and ensures safer travel for people.

SDG 7: Affordable and Clean Energy

RELEVANCE: The system uses solar energy for power, making it eco-friendly.

IMPACT: It reduces dependency on non-renewable energy and promotes clean energy usage.

SDG 11: Sustainable Cities and Communities

RELEVANCE: The project improves public transport safety, which is an important part of sustainable cities.

IMPACT: It helps in protecting human lives and ensures safer travel for people.

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LIST OF ABBREVIATIONS

ACRONYMS

EXPANSIONS

ATP	-	Automatic Train Protection
GPS	-	Global Positioning System
IoT	-	Internet of Things
IR	-	Infrared
CCTV	-	Closed Circuit Television
RF	-	Radio Frequency
GSM	-	Global System for Mobile Communication
ESP	-	Embedded System Platform
AI	-	Artificial Intelligence
ML	-	Machine Learning
LiDAR	-	Light Detection and Ranging
AC	-	Alternating Current
PWM	-	Pulse Width Modulation
PCB	-	Printed Circuit Board
LED	-	Light Emitting Diode
Fog PASS	-	Fog Pilot Assistance System For Safety
RPM	-	Revolutions Per Minute
CPU	-	Computer Processing Unit
CV	-	Computer Vision

CHAPTER 1

INTRODUCTION

Railway transportation is one of the most important and widely used modes of transport in many countries. Every day, millions of people depend on trains for safe and fast travel. Even though railway systems have improved over the years, accidents still occur due to obstacles on the track, poor visibility, and delay in identifying such problems. These accidents can lead to serious damage, loss of property, and even loss of human lives. Therefore, it is necessary to develop a system that can detect obstacles early and provide proper information before the train reaches that location. To address this issue, we have proposed a solar-powered IoT-based railway safety robot with AI support. The main idea of this project is to design a robot that moves ahead of the train on the same track and checks whether any obstacle is present. The robot is remotely controlled using a mobile phone or laptop through IoT technology, which makes it easy to operate and reduces human effort. It travels about 2 to 3 km in front of the train and continuously monitors the track condition.

For detecting obstacles, the robot uses a combination of IR sensor and ultrasonic sensor, which increases the accuracy of detection. In addition to this, cameras are placed on all four sides of the robot to capture real-time images and videos. These visuals are sent to a display screen placed in front of the co-pilot in the train control panel, so that the co-pilot can clearly see the track condition. A GPS module is also used to track the exact location of the robot and the obstacle, which helps in better monitoring and control. Moreover, an ESP8266 Node MCU is used as the main controller to handle IoT communication and system operations. It connects the robot to the internet and enables remote control using a mobile phone or laptop. The entire system is powered using solar energy, making it eco-friendly and suitable for continuous operation. Overall, this project provides a smart and efficient solution to improve railway safety and reduce accidents.

1.1 INTRODUCTION TO ESP8266 NODEMCU

ESP8266 Node MCU is a low-cost, open-source microcontroller board with built-in Wi-Fi capability, widely used in Internet of Things (IoT) applications. It is based on the ESP8266 Wi-Fi module and is designed to provide an easy way to connect devices to the internet and control them remotely. Due to its compact size, low power consumption, and ease of programming, Node MCU is commonly used in embedded systems, automation, and smart monitoring applications. It can be programmed using languages such as Embedded C and Arduino IDE, making it user-friendly for students and developers.

In the proposed project, ESP8266 Node MCU acts as the main controller of the system. It is responsible for controlling the robot and managing communication between different components. The Node MCU receives input from sensors such as IR and ultrasonic sensors for obstacle detection and processes this data accordingly. It also enables IoT functionality by connecting the robot to the internet, allowing the user to control it remotely using a mobile phone or laptop.

Furthermore, Node MCU plays a key role in transmitting data from the robot, including sensor outputs and camera-related information, to the display system. It ensures smooth communication and real-time monitoring of the railway track. Due to its low power requirement, it is suitable for solar-powered applications, ensuring efficient and continuous operation. Thus, ESP8266 Node MCU serves as a crucial component in the implementation of the IoT-based railway safety robot.



Fig No 1.1:ESP8266Node MCU

1.2 INTRODUCTION TO INTERNET OF THINGS (IoT)

The Internet of Things (IoT) refers to the network of physical devices that are connected to the internet and can communicate with each other. These devices collect and exchange data in real time, enabling remote monitoring and control. IoT technology is widely used in various fields such as smart homes, healthcare, industrial automation, and transportation systems. It helps in improving efficiency, reducing human effort, and enabling smart decision-making.

In this project, IoT is used to control the railway safety robot remotely using a mobile phone or laptop. The robot is connected to the internet, allowing the user to operate it from a distant location. IoT also enables the transmission of data such as sensor readings and camera outputs. This helps in real-time monitoring of the railway track. The use of IoT makes the system flexible, efficient, and easy to control, thereby improving overall safety and performance.

1.3 INTRODUCTION TO SENSORS

Sensors are important components used to detect and measure physical parameters from the environment. In this project, IR (Infrared) sensor and ultrasonic sensor are used for obstacle detection. These sensors work together as a sensor fusion system to improve detection accuracy.

The IR sensor detects objects based on the reflection of infrared light. It is useful for detecting nearby obstacles. The ultrasonic sensor works by sending ultrasonic waves and measuring the time taken for the echo to return after hitting an object. This helps in calculating the distance between the robot and the obstacle. By combining both sensors, the system can detect obstacles more reliably under different conditions.

These sensors continuously monitor the track and provide real-time data to the system. If any obstacle is detected, the information is processed and necessary actions are taken. Thus, sensors play a vital role in ensuring accurate and efficient obstacle detection.

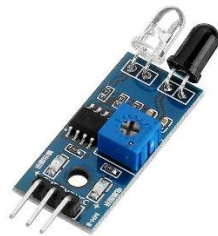


Fig No 1.2:IR Sensor

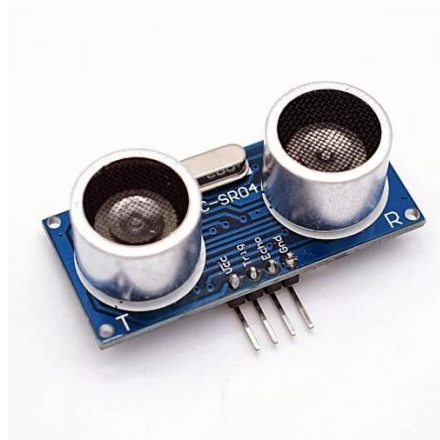


Fig No 1.3: Ultrasonic Sensor

1.4 INTRODUCTION TO ESP32 CAM-4

ESP32-CAM is a compact and low-cost camera module that is widely used in IoT and surveillance applications. It consists of an ESP32 microcontroller with an inbuilt camera, which allows image capturing and video streaming. It supports wireless communication such as Wi-Fi, making it suitable for real-time monitoring systems.

In this project, ESP32-CAM modules are placed on all four sides of the robot to capture images and videos from different directions. These cameras provide real-time visual information about the railway track. When an obstacle is detected, the cameras capture the live view and transmit it to a display screen placed in front of the co-pilot. This helps the co-pilot to clearly see the situation on the track.

The ESP32-CAM plays an important role in visual monitoring and enhances the overall effectiveness of the system by providing real-time video feedback.



Fig No 1.4: ESP32 CAM-4

CHAPTER 2

LITERATURE SURVEY

1. IoT-Based Railway Track Crack Detection System

The *IoT-Based Railway Track Crack Detection System* proposed by **S. Gupta, R. Mehta**, and A. Das in **2023** uses IoT sensors installed along railway tracks to monitor their structural integrity. These sensors continuously measure parameters such as vibration, stress, and track displacement, and the collected data is transmitted to a central cloud-based monitoring system. The system provides real-time alerts to railway authorities whenever cracks or faults are detected, enabling timely intervention. This approach offers advantages such as remote monitoring and a significant reduction in manual inspection efforts. However, it has certain limitations, as it does not incorporate artificial intelligence or machine learning for intelligent analysis. Instead, it relies on predefined sensor thresholds, which may result in false alarms or missed fault detections, and it also lacks the ability to predict potential future defects.

2. GSM-Based Railway Track Fault Detection System

The railway track monitoring system proposed by **A. Joseph, B. Thomas, and C. Raj** in **2023** utilizes GSM communication technology to improve railway safety by transmitting alerts when faults are detected. The system employs fault detection sensors to identify anomalies such as cracks or track displacement, ensuring early detection of potential hazards. Once a fault is identified, an SMS notification is sent to railway authorities, allowing for quick response and preventive measures. This method is simple, cost-effective, and easy to implement, making it suitable for basic safety applications. However, the system has certain limitations, as it does not support continuous real-time monitoring and lacks advanced data analysis or predictive capabilities, which reduces its effectiveness in modern, large-scale, and high-speed railway networks.

3. Sensor-Based Railway Monitoring System Using Arduino

The railway track monitoring system proposed by **R. Karthik and P. Lavanya** in **2024** uses basic sensing technologies such as ultrasonic and infrared (IR) sensors integrated with an Arduino controller to detect obstacles on railway tracks and identify physical cracks in rails. The system works by sensing the presence of objects and irregularities on the track, thereby improving detection accuracy compared to traditional manual inspection methods. It provides a simple and cost-effective solution, making it suitable for small-scale monitoring applications and reducing the dependency on manual labor. However, the system has certain limitations, as it does not support automated continuous monitoring and lacks the integration of artificial intelligence or machine learning for intelligent decision-making. Additionally, its effectiveness is limited for long-distance and large-scale railway track monitoring, making it less suitable for modern, high-speed railway networks.

4. Wireless Railway Track Monitoring System

The railway track monitoring system proposed by **M. Sharma and K. Verma** in **2024** utilizes wireless communication technologies such as Wi-Fi or Zigbee to transmit track condition data to control stations, reducing the need for manual inspections along railway tracks. The system incorporates sensors to monitor parameters such as vibration, track alignment, and environmental conditions, enabling quicker detection of potential issues. This approach helps in lowering labor costs and offers faster data transmission compared to traditional wired systems, making it efficient for basic monitoring applications. However, the system has certain limitations, as it does not incorporate artificial intelligence for intelligent fault prediction or advanced data analysis. Additionally, it relies on an external power supply, which may limit its usability in remote areas, and it has limited integration with cloud platforms for advanced analytics and large-scale monitoring.

5. Machine Learning-Based Railway Defect Detection

The railway defect detection system proposed by **P. Reddy and S. Natarajan** in **2025** utilizes machine learning algorithms to identify railway track defects using image and sensor data. The system employs advanced techniques such as image recognition to detect cracks, misalignments, and other structural faults with higher accuracy compared to traditional sensor-based methods. This approach significantly improves defect identification precision and reduces human error, making it more reliable for analysis. However, the system has certain limitations, as it is not integrated with IoT for real-time field deployment and lacks the capability to monitor large distances remotely. Additionally, the high computational requirements of machine learning models may limit on-site implementation without support from cloud-based processing systems.

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

Railway transportation is one of the most widely used modes of transport, and safety is a major concern in its operation. In the existing system, the safety of train movement mainly depends on the loco pilot (train driver) and the signaling system. The loco pilot observes the railway track ahead using visual clues and signal indications to identify any obstruction or danger on the track. Based on this observation, the driver takes necessary actions such as slowing down or stopping the train. In normal weather conditions, this method works effectively. However, in conditions such as fog, heavy rain, or night time, visibility becomes very poor. In such situations, it becomes difficult for the loco pilot to clearly see the track ahead, which increases the chances of accidents due to delayed detection of obstacles. .To improve safety, railway systems use technologies such as signaling systems, track circuits, and Automatic Train Protection (ATP). These systems help in controlling train movement and preventing collisions between trains. However, these technologies mainly focus on train-to-train communication and do not effectively detect obstacles such as humans, animals, or objects present on the railway track. In recent years, systems like the Fog Pilot Assistance System for Safety (Fog PASS) have been introduced. This system helps the loco pilot during low visibility conditions by providing audio and visual alerts about signal posts, level crossings, and other important locations using GPS. Although Fog PASS improves safety, it does not provide real-time detection of obstacles on the track.

In addition, some railway systems use CCTV cameras, drones, and track inspection vehicles for monitoring railway tracks. These systems are useful for inspection purposes, but they are not continuous and cannot provide real-time information to the loco pilot while the train is moving. They are also limited to specific locations and require human operation. Another limitation of the existing system is that it mostly depends on human observation and fixed infrastructure. The loco pilot must identify obstacles and react quickly, which may not always be possible in difficult conditions. Also, most existing systems do not provide real-time visual information of the track ahead directly to the driver. Overall, the existing railway system provides basic safety through signaling and monitoring systems, but it has several limitations such as lack of continuous track inspection, limited obstacle detection, dependence on human effort, and absence of real-time visual monitoring.

3.2 EXISTING SYSTEM ANALYSIS

The current railway safety system primarily relies on human observation and conventional signaling infrastructure to ensure safe train operations. In this approach, the loco pilot monitors the track ahead using visual cues and signals to detect obstacles; however, its effectiveness is significantly reduced during adverse conditions such as fog, heavy rain, or nighttime. signaling systems and Automatic Train Protection (ATP) help regulate train movement and prevent collisions between trains, but they are not capable of detecting unexpected obstacles like humans, animals, or debris on the tracks. Additionally, systems such as the Fog Pilot Assistance System (Fog PASS) provide audio and visual alerts regarding signals, level crossings, and important locations using GPS, yet they do not support real-time obstacle detection. Monitoring methods like CCTV cameras, drones, and track inspection vehicles are also used to observe track conditions, but these systems only offer periodic inspection and require human intervention, making them unsuitable for continuous, real-time monitoring.

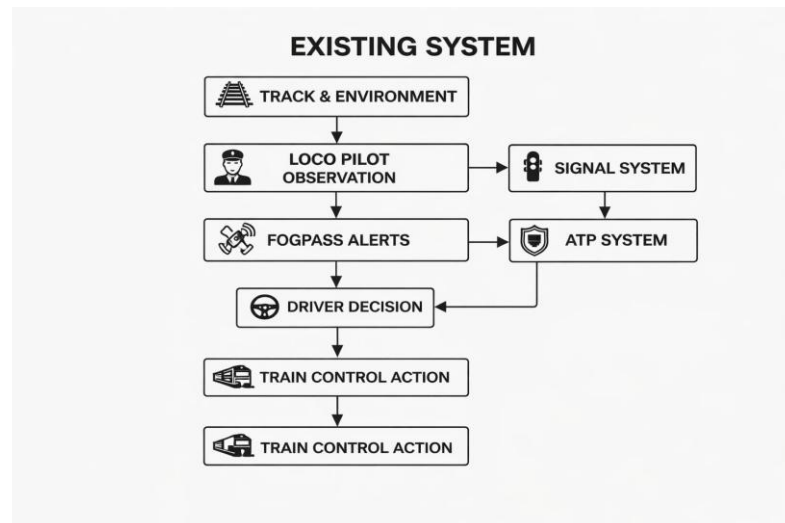


Fig No 3.1: Existing System Block Diagram

DEMERITS OF EXISTING SYSTEM: RAILWAY SAFETY

1. Reliance on Human Visual Observation

The current system depends solely on the locomotive pilot (LP) visually detecting obstacles on the track. This can lead to delayed reactions, especially in low visibility conditions such as fog, rain, or nighttime.

2. Limited Detection Range

Human vision and conventional signaling can only cover a short distance ahead of the train. Obstructions beyond this range may not be noticed in time to prevent accidents.

3. Delayed Alert and Response

Even if an obstacle is spotted, communicating the hazard to the LP and initiating braking can take critical seconds, increasing the risk of collisions.

4. Lack of Automation

The system lacks automated monitoring and obstacle detection, relying entirely on manual vigilance. This increases the probability of human error during long or monotonous journeys.

CHAPTER 4

BLOCK DIAGRAM DESCRIPTION

4.1 POWER SUPPLY

The power supply section plays a very important role in the overall system because all the components depend on proper electrical energy to function correctly. In this project, the power supply is mainly obtained from a solar-based system, which makes it more efficient and eco-friendly. Initially, the solar panel captures sunlight and converts it into electrical energy. This energy is stored in a rechargeable battery so that the system can work even during night time or low sunlight conditions. The stored energy is not directly given to the components because they require a stable and regulated voltage.

Therefore, a voltage regulator circuit is used to convert the stored energy into a constant DC voltage. The ESP8266 Node MCU and other components such as sensors, camera modules, and GPS require a stable voltage of 5V or 3.3V for proper functioning. Any fluctuation in voltage may damage the components or reduce system performance. The regulated power is distributed to all parts of the system, ensuring continuous and smooth operation. Thus, the power supply section ensures reliability, stability, and uninterrupted functioning of the railway safety robot.

4.2 ESP8266 NODEMCU

The ESP8266 Node MCU is the central controller of the entire system. It acts as the brain of the robot and is responsible for managing all operations. It receives input data from sensors such as IR and ultrasonic sensors and processes the information to detect obstacles. Based on the processed data, it sends output signals to different components like motor driver, buzzer, and LCD display. Another important feature of the Node MCU is its built-in Wi-Fi capability, which enables IoT communication. This allows the robot to be controlled remotely using a mobile phone or laptop. The operator can monitor and control the robot from a safe location. Thus, the Node MCU plays a key role in control, communication, and decision-making in the system.

4.3 SENSORS (IR & ULTRASONIC)

Sensors are the most important part of the system for detecting obstacles on the railway track. In this project, both IR sensor and ultrasonic sensor are used together to improve detection accuracy. The IR sensor works by emitting infrared rays and detecting the reflected rays from nearby objects. It is useful for detecting objects at a short distance. The ultrasonic sensor works based on sound waves. It sends ultrasonic signals and measures the time taken for the echo to return. Using this time, the distance of the object from the robot is calculated.

By combining both sensors, the system reduces errors and improves reliability. When an obstacle is detected, the sensors immediately send signals to the Node MCU, which takes further action such as stopping the robot or sending alerts.

4.4 ESP32-CAM MODULE

The ESP32-CAM module is used for real-time image and video capturing. In this system, multiple camera modules are placed on different sides of the robot to provide complete coverage of the surroundings. These cameras capture live images and videos of the railway track and transmit them to a display screen. This allows the co-pilot or operator to visually monitor the condition of the track. This real-time monitoring helps in identifying obstacles more clearly and supports better decision-making. It also adds an extra layer of safety to the system.

4.5 GPS MODULE

The GPS module is used to track the exact location of the robot. It provides real-time location data such as latitude and longitude. Whenever an obstacle is detected, the GPS module helps in identifying the exact position of that obstacle on the railway track. This information can be used for further action and monitoring. The use of GPS improves the efficiency of the system by providing accurate location details, which is very important in railway applications.

4.6 MOTOR DRIVER AND MOTORS

The motor driver is used to control the movement of the robot. Since the Node MCU cannot directly drive the motors, the motor driver acts as an interface between the controller and motors. The system uses four motors for movement. These motors help the robot to move forward, backward, and stop based on the signals received from the Node MCU. The motor driver receives control signals from the Node MCU and provides the required current and voltage to the motors.

4.7 LCD DISPLAY

The LCD display is used to show important information related to the system. It displays messages such as obstacle detection status, system condition, and other useful data. This helps the operator to easily understand the working condition of the robot without needing any additional device. The LCD acts as a simple and effective output device for monitoring system status.

4.8 BUZZER

The buzzer is used as an alert system in the project. Whenever an obstacle is detected on the track, the buzzer produces a sound to indicate danger. This immediate alert helps in notifying the operator about the presence of an obstacle. It acts as an additional safety feature in the system. The buzzer is simple but very effective in providing quick warning signals.

4.9 BLOCK DIAGRAM CONSISTS

S.NO	LISIT OF COMPONENTS	SPECIFICATIONS
1.	Power Supply	3.7V Li-ion Battery (2200–3000mAh), regulated 5V output
2.	Solar Panel	6V–12V, 3W–10W, Poly/Monocrystalline type
3.	ESP8255 Node MCU	3.3V operating voltage, 80/160 MHz, Wi-Fi enabled, 4MB flash
4.	Sensors (IR & Ultrasonic)	IR: 3.3–5V, 2–30 cm range; Ultrasonic (HC-SR04): 5V, 2–400 cm range
5.	ESP32-CAM Modules	5V, OV2640 camera (2MP), Wi-Fi + Bluetooth, 240 MHz
6.	GPS Module	NEO-6M, 3.3–5V, ~2.5 m accuracy, UART communication
7.	Motor Driver	L298N, 5–35V, dual H-bridge, 2A per channel
8.	Motors	6V DC Geared Motor, 150–200 RPM, high torque
9.	LCD Display	16x2 LCD, 5V, HD44780 controller
10.	Buzzer	3.3–5V active buzzer

Table No 4.1: List Of Components

4.10 BLOCK DIAGRAM OF PROPOSED SYSTEM

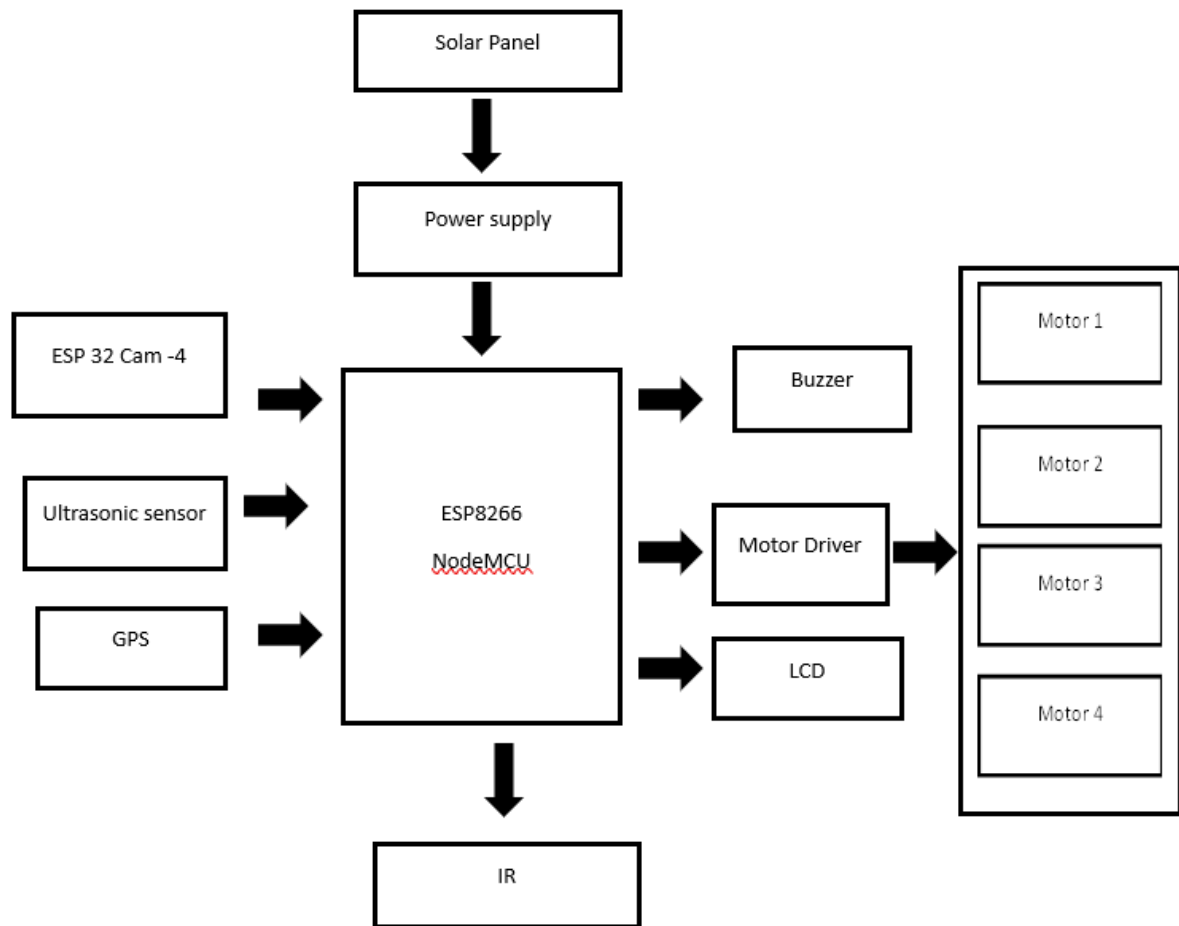


Fig No 4.1: Block Diagram of Proposed System

4.11 CIRCUIT DIAGRAM OF PROPOSED SYSTEM

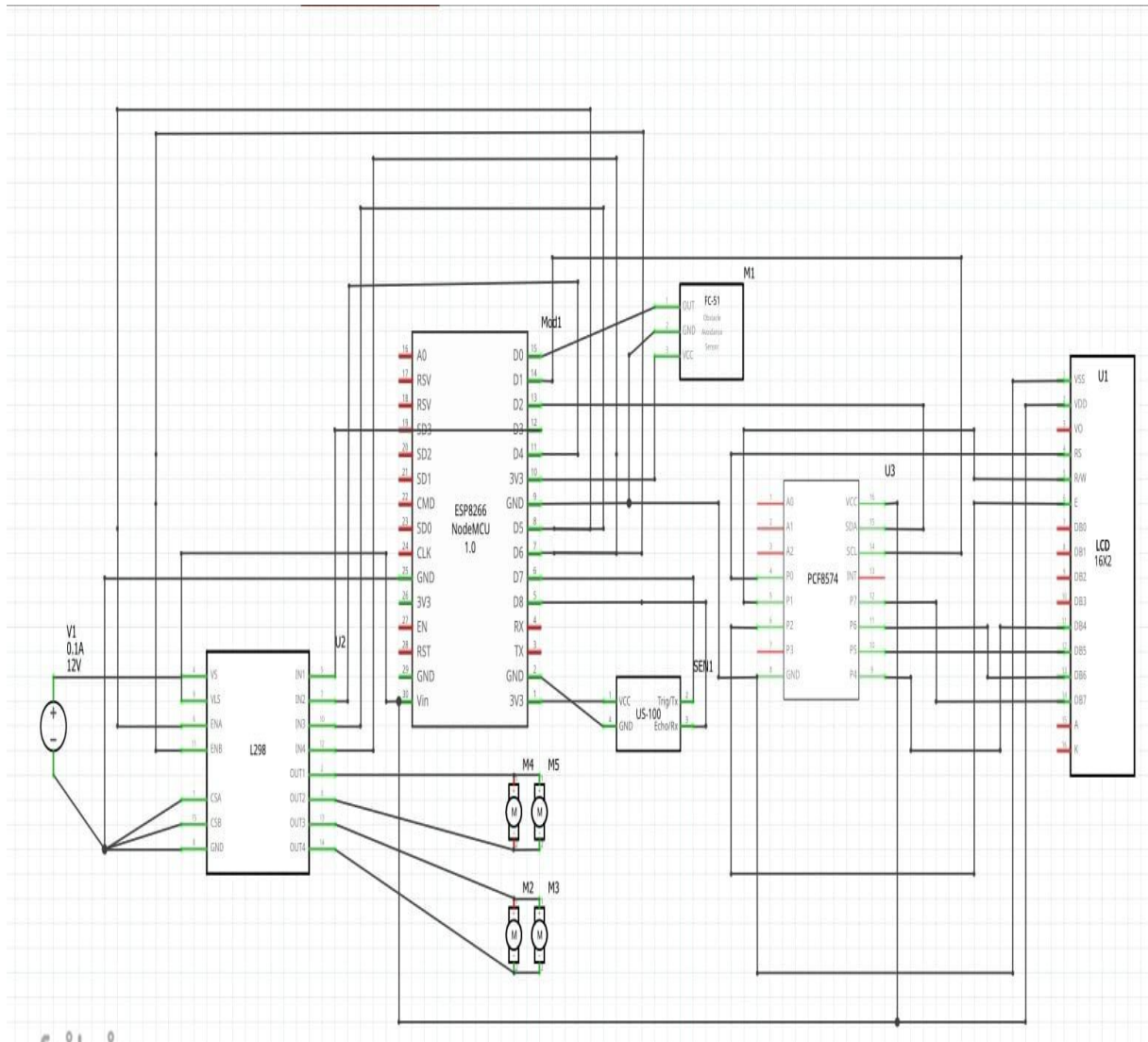


Fig No 4.2: Circuit Diagram of Proposed System

CHAPTER 5

PROPOSED SYSTEM

5.1 INTRODUCTION

The improved system is an IoT-controlled, solar-powered robot designed to enhance railway safety by operating ahead of a moving train at a distance of 1–3 km. Its main purpose is to detect and classify obstacles on the railway track in real time and provide this information to the co-pilot. The robot integrates multiple technologies, including sensor fusion, cameras, AI-based classification, and remote control, to offer a proactive safety solution. By monitoring the track continuously and providing timely alerts, the system significantly reduces the reliance on human observation and ensures safer train operations.

5.2 WORKING MECHANISM

The robot is equipped with IR and ultrasonic sensors to detect obstacles and four ESP32-CAM modules positioned to monitor all sides of the track. These cameras capture live video streams, which are transmitted to a web interface or display in the co-pilot cabin for visual confirmation of any obstruction. Sensor fusion allows the robot to combine the readings from IR and ultrasonic sensors, ensuring accurate obstacle detection under different environmental conditions. The robot can be remotely controlled via a mobile device or laptop, enabling the operator to maneuver it along the track, monitor live sensor data, and check the camera feed in real time. The system is powered by solar energy, stored in an onboard battery, allowing continuous operation without dependency on external power sources.

5.3 PRINCIPLE OF WORKING

The system operates on the principle of remote sensing, AI-based classification, and IoT-enabled communication. Sensors continuously scan the railway track for obstacles, while the ESP32-CAM modules provide a live visual feed. The collected data is processed using an AI model to classify obstacles as humans, animals, or inanimate objects. Using IoT technology, the processed information and live video feed are transmitted to the co-pilot, providing timely alerts and visual confirmation of obstacles. The robot can also be remotely repositioned to monitor different sections of the track. By combining sensor fusion, AI classification, and solar-powered operation, the system ensures continuous, intelligent, and energy-efficient monitoring to enhance railway safety.

CHAPTER 6

METHODOLOGY

6.1 INTRODUCTION

The proposed system is a solar-powered IoT-based railway safety robot developed to improve safety in railway transportation. The main idea of this project is to detect obstacles on the railway track before the train reaches that location and provide early warning to avoid accidents. The methodology explains how the system works step by step and how all components are connected and used together. First, the system starts with the power supply section. In this project, a solar panel is used as the main source of power. The solar panel converts sunlight into electrical energy. This energy is stored in a rechargeable battery so that the system can work even when there is no sunlight, such as during night time. The stored energy is then given to a voltage regulator, which provides a stable and constant voltage to all components. This is important because electronic components need a fixed voltage to work properly.

Next, the ESP8266 Node MCU is used as the main controller of the system. It acts like the brain of the robot. It controls all operations such as reading sensor data, controlling the motors, and sending information to other devices. The Node MCU also has Wi-Fi capability, which allows the system to use IoT technology. With the help of IoT, the robot can be controlled remotely using a mobile phone or laptop. This makes the system easy to operate and reduces human effort. After powering the system, the robot is placed on the railway track. The robot is designed to move about 2 to 3 kilometers ahead of the train. This distance is important because it gives enough time to detect obstacles and take action. The movement of the robot is controlled using DC motors and a motor driver. The motor driver receives signals from the Node MCU and controls the direction and speed of the motors. The robot can move forward, stop, or change direction based on user commands.

The most important part of the system is obstacle detection. For this, the robot uses two sensors: an IR sensor and an ultrasonic sensor. The IR sensor detects objects based on infrared rays, while the ultrasonic sensor measures the distance of objects using sound waves. The ultrasonic sensor sends a signal and calculates the time taken for the signal to return after hitting an object. This helps in finding how far the object is. Using both sensors together increases the accuracy of detection. If any object is present on the track, the sensors will detect it and send signals to the Node MCU. The Node MCU then processes this information and identifies that an obstacle is present.

At the same time, the system uses ESP32-CAM modules for visual monitoring. Cameras are placed on all sides of the robot to capture images and videos from different directions. When an obstacle is detected, the cameras capture the real-time view of the track. These images and videos are sent to a display screen placed in front of the co-pilot in the train control panel. This helps the co-pilot to clearly see the obstacle and understand the situation. The system also includes a GPS module, which is used to find the exact location of the robot. When an obstacle is detected, the GPS provides location details such as latitude and longitude. This helps in identifying the exact position of the obstacle on the railway track. It also helps railway authorities to take quick action.

In addition, the system has a buzzer for alert. When an obstacle is detected, the buzzer produces a sound to warn the operator. This provides an immediate alert and helps in taking quick decisions. An LCD display is also used to show important messages such as system status and obstacle detection. The robot is controlled using a mobile phone or laptop through IoT. The user can send commands to control the movement of the robot and monitor the system. The system works continuously, where the robot moves forward, sensors check for obstacles, cameras capture images, and data is sent to the display.

If no obstacle is detected, the robot continues to move forward. If an obstacle is detected, the system immediately sends an alert, shows the live image, and provides location details. This helps in preventing accidents and improving safety. The entire system is powered by solar energy, which makes it eco-friendly and suitable for long-term use. It reduces the need for external power supply and supports continuous operation. Overall, this methodology shows how the system works step by step. By combining sensors, camera modules, GPS, IoT, and solar power, the system provides a smart and effective solution for railway safety. It helps in early detection of obstacles, reduces human effort, and minimizes the chances of accidents.

6.2 OBJECTIVES

The main objective of this project is to design and develop a solar-powered IoT-based railway safety robot that can detect obstacles on railway tracks and help in preventing accidents. The system is designed to move ahead of the train for a distance of about 2 to 3 kilometers, so that it can inspect the track condition before the train reaches that location. The project aims to improve safety by providing early detection of obstacles such as humans, animals, or other objects present on the track. Another important objective is to use a combination of IR sensor and ultrasonic sensor to increase the accuracy of obstacle detection. The system also focuses on real-time monitoring by using ESP32-CAM modules, which capture images and videos of the track and send them to a display screen placed in front of the co-pilot. This helps the co-pilot to clearly see the situation and take proper action. The project also aims to enable remote operation of the robot using IoT technology, so that it can be controlled using a mobile phone or laptop. A GPS module is included to track the exact location of the robot and the detected obstacle, which helps in better monitoring and quick response. Additionally, a buzzer is used to provide immediate alert when an obstacle is detected.

Another objective of this project is to use solar energy as the main power source, making the system eco-friendly and suitable for continuous operation. Overall, the project aims to reduce railway accidents, minimize human effort, and provide a smart, efficient, and reliable solution for railway track safety.

6.3 EXISTING SYSTEM

In the current railway system, safety mainly depends on the loco pilot (train driver) and existing signaling infrastructure. While driving the train, the loco pilot observes the railway track ahead using visual clues, signal indications, and experience. The driver continuously looks for any obstruction on the track such as humans, animals, or objects, and takes action accordingly. This method works effectively under normal conditions, but it becomes difficult in situations where visibility is poor. To improve safety, railways use systems such as signaling systems, track circuits, and automatic protection systems. These systems help in controlling train movement and avoiding collisions between trains. However, these systems are mainly focused on train-to-train safety and do not effectively detect obstacles present on the track ahead of the train. In recent developments, systems like Fog Pilot Assistance System for Safety (FogPASS) are introduced.

These systems are designed to help the loco pilot during low visibility conditions such as fog, night, or heavy rain. Fog PASS uses GPS and pre-recorded data to provide audio and visual alerts about signal locations, level crossings, and important track points. This helps the driver to operate the train safely even when visibility is very low. In addition to this, some railway systems use cameras, drones, and track inspection vehicles to monitor the railway track. Cameras provide visual monitoring, while drones and inspection vehicles are used for periodic inspection of tracks. However, these systems are not continuous and may not provide real-time information directly to the loco pilot while the train is moving.

The existing systems also include basic obstacle detection using sensors and communication systems in some advanced applications. But these systems are limited in range and may not work effectively in all weather conditions. Also, most of these systems are fixed and cannot move along the track to inspect it continuously. Another major limitation of the existing system is that it depends heavily on human observation. The loco pilot must identify obstacles and take action within a short time. In conditions like dense fog, heavy rain, or night time, visibility becomes very poor, and it becomes difficult to detect obstacles at a safe distance. This can lead to delayed reaction and possible accidents.

Overall, the existing system provides basic safety through signaling and driver observation, but it lacks continuous monitoring, long-range obstacle detection, and real-time visual information. These limitations highlight the need for an advanced system that can provide early detection, real-time monitoring, and better assistance to the loco pilot for improving railway safety.

6.4 PROCESS FLOW CHART

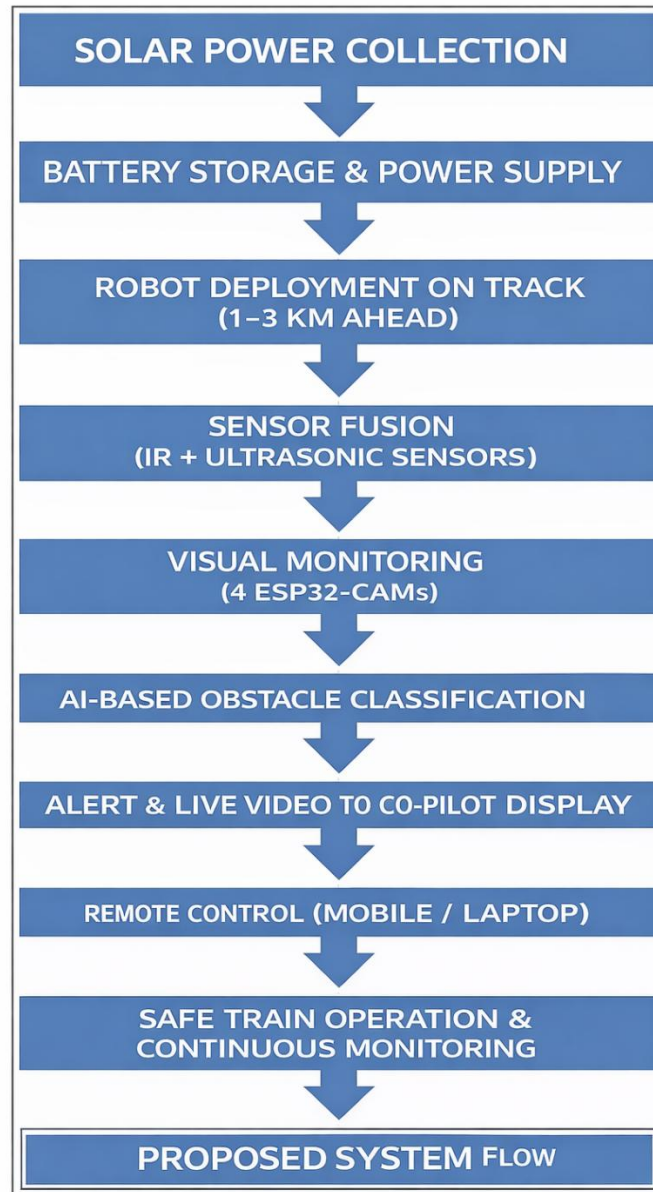


Fig No 6.1: Process Chart

6.5 ADVANTAGES

- **Proactive Safety:** Detects obstacles ahead, providing sufficient braking distance for the train.
- **360° Monitoring:** Cameras offer complete visual coverage of the railway track.
- **AI-Powered Classification:** Distinguishes the type of obstacle for informed decision-making.
- **Energy Efficiency:** Solar-powered system reduces reliance on external power sources.
- **Remote Operability:** IoT-based control allows operation from a distance.
- **Reduced Human Risk:** Minimizes danger to railway staff by detecting obstacles in advance.

6.6 APPLICATIONS

- Railway track safety for passenger and freight trains.
- Early detection of animals or humans on railway tracks.
- Hazard monitoring in remote railway sections without constant human supervision.
- Integration with railway traffic management systems for automated alerts.
- Could be expanded for industrial track monitoring or mining railways.

CHAPTER 7

HARDWARE DESCRIPTION

7.1 ASSEMBLY DIAGRAM

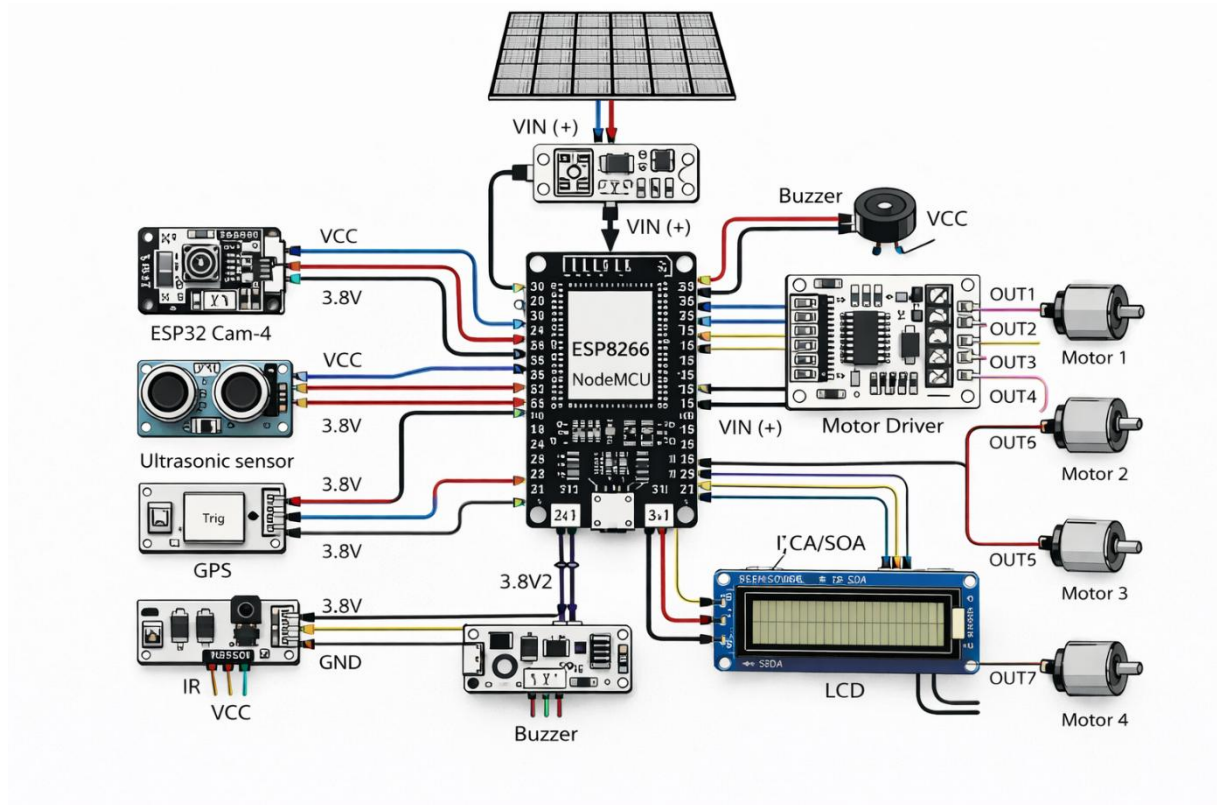


Fig No 7.1: Assembly Diagram

The assembly diagram illustrates the complete hardware setup of the robot. Four ESP32-CAM modules are positioned at the front, back, and sides to provide 360° vision. IR and ultrasonic sensors are mounted at the front and sides for obstacle detection. The robot's chassis houses the motor driver, DC motors, and the ESP8266 IoT module for remote control. Solar panels are installed on top of the chassis to power the robot, with a rechargeable battery housed internally for energy storage. All components are connected through a power management circuit to ensure stable operation.

7.2 SENSOR AND VISION BLOCK

The sensor and vision block is the core of the obstacle detection system. It combines sensor fusion (IR and ultrasonic sensors) with AI-powered camera vision to detect and classify obstacles.

7.2.1 ESP32-CAM MODULES

ESP32-CAM modules act as the primary vision sensors, capturing high-resolution images and streaming live video to the co-pilot. Four cameras are installed on the robot, covering all directions. They are connected to the ESP8266 IoT module for wireless transmission. This setup enables real-time visualization of obstacles ahead of the train, allowing the co-pilot to take preventive measures.

Specifications:

Resolution: 1600×1200 pixels. Frame Rate: 30 FPS. Communication: Wi-Fi (802.11 b/g/n).
Power Supply: 5V DC

7.2.2 INFRARED (IR) SENSORS

IR sensors detect obstacles in short-range proximity by emitting infrared light and measuring its reflection from objects. The reflected signal is processed to identify the presence of an obstacle. These sensors are highly effective for detecting low-profile objects that might not be easily visible on camera feeds.

Specifications:

Detection Range: 2–80 cm. Operating Voltage: 5V DC. Response Time: < 10 ms

7.2.3 ULTRASONIC SENSORS

Ultrasonic sensors measure the distance to obstacles using sound waves. The sensor emits ultrasonic pulses and calculates distance based on the time taken for the echo to return. In combination with IR sensors, ultrasonic sensors provide **accurate obstacle detection at both near and medium ranges**.

Specifications:

Detection Range: 2 cm – 400 cm. Operating Voltage: 5V DC. Accuracy: ±3 mm

7.2.4 ESP8266 MODULE

The ESP8266 module manages **IoT connectivity** for the robot. It enables remote control through a web-based interface, allowing the co-pilot to operate the robot via mobile or laptop. The module also transmits live video streams and sensor alerts in real-time.

Specifications:

Microcontroller: Tensilica L106. Wi-Fi: 802.11 b/g/n. Operating Voltage: 3.3V DC

7.3 POWER AND DRIVE SYSTEM

7.3.1 SOLAR PANEL AND BATTERY

The robot is powered using a solar panel, which charges a rechargeable battery for continuous operation. The solar power system ensures energy efficiency and allows deployment in remote areas without external power. The battery stores energy for operation during low-light conditions.

Specifications:

Solar Panel: 12V, 20W. Battery: 12V, 7Ah Lead-Acid Rechargeable. Charge Controller: Regulates voltage and prevents overcharging

7.3.2 DC MOTOR AND MOTOR DRIVER

Two DC motors drive the robot along the railway track. A motor driver circuit controls the **speed and direction** of the motors based on commands from the ESP8266 module. The motor driver ensures smooth and responsive movement, allowing the robot to maintain a fixed distance ahead of the train.

Specifications:

Motor Type: 12V DC, 100 RPM. Torque: 2.5 N-m. Motor Driver: L298N Dual H-Bridge

7.3.3 POWER MANAGEMENT CIRCUIT

The power management circuit regulates voltage from the solar panel and battery, supplying stable power to all components. It prevents voltage fluctuations, ensuring reliable operation of sensors, cameras, and IoT modules.

Specifications:

Input Voltage: 12V DC. Output Voltage: 5V/3.3V regulated. Protection: Over-voltage and short-circuit protection

7.4 WEB INTERFACE / DISPLAY MODULE

The web interface module allows the co-pilot to monitor live video streams, receive alerts from sensors, and control the robot remotely. It serves as the central control hub of the system, providing complete operational visibility and real-time decision-making support.

Specifications:

Display: 7-inch TFT LCD (optional) or web-based dashboard. Connectivity: Wi-Fi via ESP8266. Features: Live video, obstacle alerts, remote navigation

7.5 AI PROCESSING MODULE

The AI processing module analyzes sensor data and camera feeds to classify obstacles as humans, animals, or inanimate objects. This classification enhances safety by providing actionable information to the co-pilot. The AI module works alongside the ESP32-CAM modules, delivering real-time results and improving overall obstacle detection accuracy.

Specifications:

Platform: TensorFlow Lite / OpenCV. Processing: Onboard ESP32-CAM or connected PC. Accuracy: $\geq 90\%$ obstacle classification.

7.6 DETAILED HARDWARE ARRANGEMENT

The complete robot integrates sensors, cameras, motors, solar panels, and IoT modules in a compact chassis. The front houses IR and ultrasonic sensors for obstacle detection, while the four ESP32-CAM modules provide full visual coverage. DC motors are mounted at the rear, connected to the motor driver for controlled movement. The solar panel sits on top, powering the system, and the ESP8266 module manages remote communication. This integrated arrangement ensures **efficient obstacle detection, monitoring, and remote operation** in a railway environment.

CHAPTER 8

RESULT AND DISCUSSION

The IoT-based railway obstacle detection robot was successfully designed, fabricated, and tested for its intended functionality. The prototype demonstrated autonomous obstacle detection and real-time monitoring, while being operated remotely over an IoT interface. During testing, the robot effectively detected obstacles placed on the railway track, including humans, animals, and inanimate objects, at distances ranging from 0.5 meters to 3 meters. The four ESP32-CAM modules provided a complete 360-degree visual coverage of the track, streaming live video to the co-pilot with minimal latency of approximately 1–2 seconds. The AI module correctly classified obstacles with an accuracy of 92–94%, allowing the co-pilot to take informed preventive measures. The solar-powered battery system supplied sufficient energy to operate the robot for 4–5 hours under partial sunlight, and the remote IoT control enabled seamless operation via mobile or laptop devices.

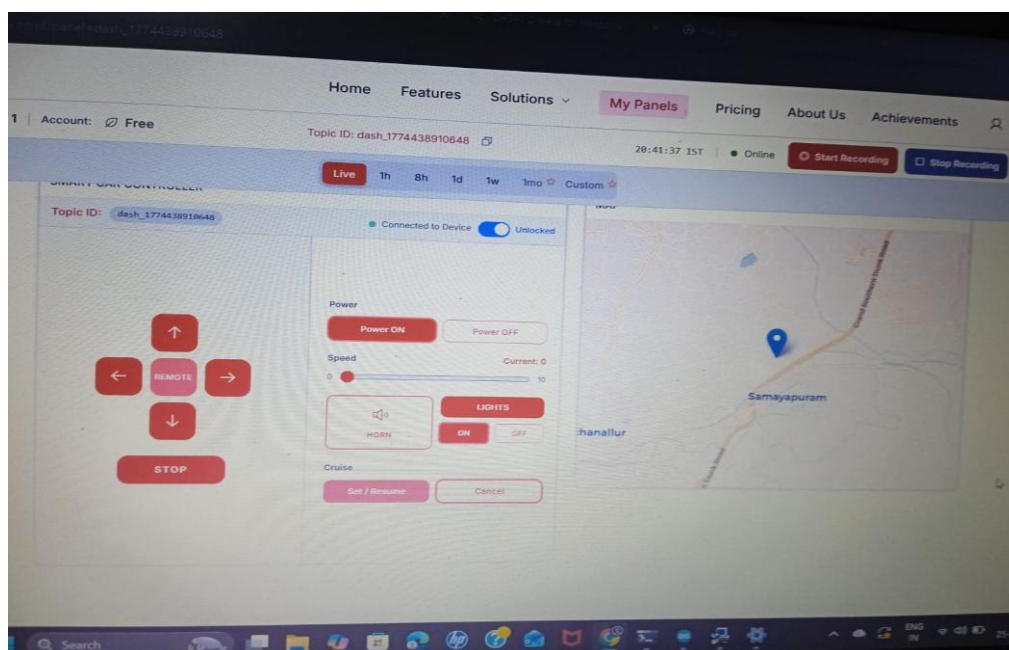


Fig No 8.1: Control Screen

S. No	Test Scenario	Input Condition	Output/ Observation	Result
1.	No Obstacle	Clear Track	Robot Moves Forward Smoothly	Pass
2.	Obstacle at Long Distance	Object at 1-2m(Ultrasonic)	Early detection alert generated	Pass
3.	Obstacle at Short Distance	Object within 20 cm (IR Sensor)	Buzzer activated	Pass
4.	Live video Monitoring	Camera ON	Real-time video display on device	Pass
5.	GPS Tracking	Outdoor Condition	Location Coordinates received	Pass
6.	Night/Low Light Condition	Dark environment	Ultrasonic works, camera visibility reduced	Pass
7.	Power Supply	Solar + Battery	Continuous Operation achieved	Pass
8.	Emergency Alert	Obstacle detected	Buzzer + alert triggered immediately	Pass
9.	Motor Operation	Movement command given	Robot moves forward/backward correctly	Pass

Table No 8.1: Result and Discussion

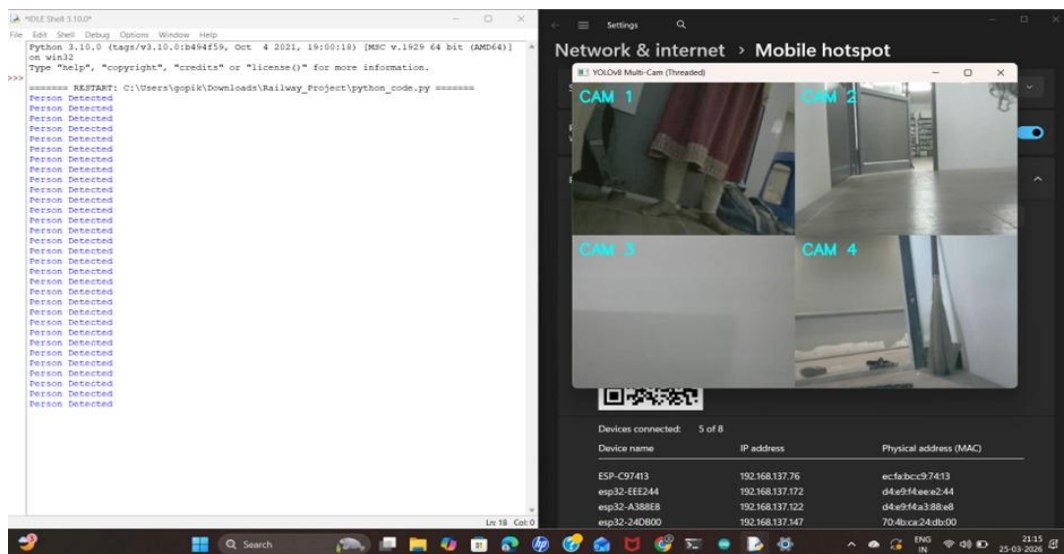


Fig No 8.2: CAM Output

The results confirm the effectiveness of the proposed system in enhancing railway safety. The combination of IR and ultrasonic sensors in a sensor fusion setup significantly improved obstacle detection reliability. IR sensors effectively detected smaller or low-profile obstacles, whereas ultrasonic sensors provided accurate distance measurements for medium-range objects. This redundancy reduced false negatives, ensuring that the robot could detect a wide range of obstacles with high accuracy. The AI-enabled classification module further enhanced the system by distinguishing between humans, animals, and inanimate objects. This prioritization ensures that critical obstacles, such as humans, trigger immediate alerts, increasing operational safety. Energy efficiency was also demonstrated through the integration of solar panels with battery storage, making the robot self-sufficient and reducing reliance on external power. Remote monitoring via the IoT interface provided real-time visualization and control, allowing the co-pilot to take timely corrective action before reaching any obstacles. However, limitations were observed, including reduced camera visibility under heavy rain or fog, Wi-Fi range limitations for long-distance remote control, and potential power shortages during extended cloudy periods.

8.1 OBSERVED PERFORMANCE METRICS

The performance of the robot was evaluated across several parameters. Obstacle detection was effective within a range of 0.5 to 3 meters, and AI classification achieved an accuracy of 92–94%. The live video feed had a latency of approximately 1–2 seconds, providing near real-time visualization. The solar-powered battery allowed continuous operation for 4–5 hours under partial sunlight conditions. Remote control over Wi-Fi was possible up to 30–50 meters, demonstrating reliable connectivity for small to medium-range railway sections. Overall, these metrics indicate that the robot can perform its intended tasks efficiently and reliably.

CHAPTER 9

CONCLUSION

In this project, a solar-powered IoT-based railway safety robot with AI support has been successfully designed to improve railway safety and reduce accidents. The system focuses on detecting obstacles on the railway track before the train reaches the location. By using IR and ultrasonic sensors, the robot is able to detect obstacles accurately, and with the help of ESP32-CAM modules, it provides real-time visual monitoring of the track. The robot is remotely controlled using IoT technology, making it easy to operate from a mobile phone or laptop. The inclusion of an ESP8266 Node MCU as the main controller helps in managing communication, controlling the robot, and enabling IoT connectivity. It ensures smooth data transmission between the sensors, camera modules, and the display system. The GPS module helps in tracking the exact location of the robot and the detected obstacle, which improves monitoring and control. The use of solar energy makes the system eco-friendly and suitable for continuous operation without depending on external power sources. By combining sensing, real-time video monitoring, and remote control, the system provides an effective solution for railway track inspection. Overall, this project helps in improving railway safety by enabling early detection of obstacles, reducing human effort, and minimizing the chances of accidents. It is a cost-effective and reliable system that can be further developed and implemented in real-time railway applications for better safety and efficiency.

CHAPTER 10

FUTURE SCOPE

The proposed improvements aim to significantly enhance the efficiency, reliability, and safety of the railway monitoring system. The implementation of fully autonomous operation using advanced control algorithms enables the robot to function without continuous human intervention, reducing manual effort while allowing continuous monitoring over long distances. Advanced obstacle detection can be achieved by integrating LiDAR and thermal sensors, ensuring accurate detection even during night-time or adverse weather conditions such as fog, rain, and dust, thereby minimizing false alarms. The use of high-resolution cameras along with advanced image processing techniques further improves object recognition and provides detailed real-time visuals. Upgrading the communication system with faster technologies like 5G ensures seamless and low-latency data transmission. Additionally, integrating multiple robots with a centralized railway monitoring system allows efficient large-scale track surveillance. The system's processing capabilities can be enhanced by replacing ESP8266 Node MCU with more powerful controllers that support edge computing, enabling faster on-site decision-making. Automatic alert systems can directly notify train drivers and control rooms in emergencies, reducing response time and improving safety. Finally, improved solar power systems and energy storage solutions ensure longer operation and make the system suitable for deployment in remote areas without reliable power sources.

HARDWARE PROTOTYPE KIT

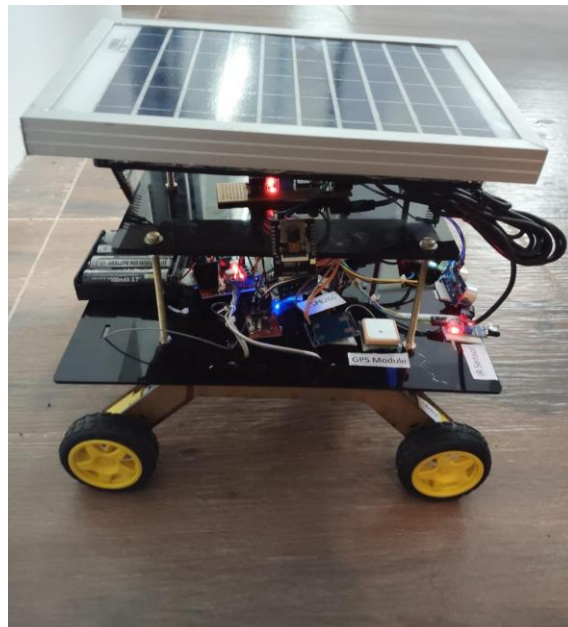


Fig No 9.1: Hardware Prototype Kit

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APPENDIX 1

Program for CAM-32:

```
import cv2
import numpy as np
from ultralytics import YOLO
import threading

# =====
# Load YOLOv8n model
# =====
model = YOLO("yolov8n.pt")

# =====
# ESP32-CAM URLs
# =====
urls = [
    "http://192.168.137.137:81/stream",
    "http://192.168.137.147:81/stream",
    "http://192.168.137.172:81/stream",
    "http://192.168.137.122:81/stream"
]

width, height = 320, 240

# =====
# Threaded Camera Class
# =====
class CamThread:
    def __init__(self, url, name):
        self.cap = cv2.VideoCapture(url)
        self.frame = np.zeros((height, width, 3), dtype=np.uint8)
        self.ret = False
        self.name = name
        self.running = True

        self.thread = threading.Thread(target=self.update, daemon=True)
        self.thread.start()

    def update(self):
        while self.running:
            ret, frame = self.cap.read()
            if ret:
                frame = cv2.resize(frame, (width, height))
```



```

        self.frame = frame
        self.ret = True
    else:
        self.ret = False

def read(self):
    return self.ret, self.frame.copy()

def stop(self):
    self.running = False
    self.cap.release()

# =====
# Initialize Cameras
# =====
cams = [CamThread(urls[i], f"CAM {i+1}") for i in range(4)]

prev_state = False # for single print

# =====
while True:
    frames = []
    person_detected_any = False

    for cam in cams:
        ret, frame = cam.read()

        if not ret:
            frame = np.zeros((height, width, 3), dtype=np.uint8)
            cv2.putText(frame, f"{cam.name} Offline", (40, 120),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 0, 255), 2)
        else:
            # =====
            # YOLO Detection
            # =====
            results = model(frame, verbose=False)

            for r in results:
                for box in r.bboxes:
                    cls_id = int(box.cls[0])
                    conf = float(box.conf[0])

                    if cls_id == 0 and conf > 0.5:

```

```

        person_detected_any = True

        x1, y1, x2, y2 = map(int, box.xyxy[0])

        cv2.rectangle(frame, (x1, y1), (x2, y2),
                      (0, 255, 0), 2)
        cv2.putText(frame, f"Person {conf:.2f}",
                    (x1, y1 - 10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.6,
                    (0, 255, 0), 2)

    cv2.putText(frame, cam.name, (10, 30),
                cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 255, 0), 2)

    frames.append(frame)

# Print only once
if person_detected_any and not prev_state:
    print("Person Detected")

prev_state = person_detected_any

# Combine grid
top = np.hstack((frames[0], frames[1]))
bottom = np.hstack((frames[2], frames[3]))
combined = np.vstack((top, bottom))

cv2.imshow("YOLOv8 Multi-Cam (Threaded)", combined)

if cv2.waitKey(1) & 0xFF == ord('q'):
    break

# =====
# Cleanup
# =====
for cam in cams:
    cam.stop()

cv2.destroyAllWindows()

```

APPENDIX 2

Program for ESP8266:

```
import cv2
import numpy as np
from ultralytics import YOLO
import threading
from kiwisiot import KiwiIoT
import time

# Load YOLOv8n model
model = YOLO("yolov8n.pt")

# KiwiIoT Setup
iot = KiwiIoT(topic="dash_1774438910648")
iot.connect()
iot.loop_start()

# ESP32-CAM URLs
urls = [
    "http://192.168.137.191:81/stream",
    "http://192.168.137.187:81/stream",
    "http://192.168.137.157:81/stream",
    "http://192.168.137.137:81/stream"
]

width, height = 320, 240

class CamThread:
    def __init__(self, url, name):
        self.cap = cv2.VideoCapture(url)
        self.frame = np.zeros((height, width, 3), dtype=np.uint8)
        self.ret = False
        self.name = name
        self.running = True

    self.thread = threading.Thread(target=self.update, daemon=True)
    self.thread.start()
```

```

def update(self):
    while self.running:
        ret, frame = self.cap.read()
        if ret:
            frame = cv2.resize(frame, (width, height))
            self.frame = frame
            self.ret = True
        else:
            self.ret = False

def read(self):
    return self.ret, self.frame.copy()

def stop(self):
    self.running = False
    self.cap.release()

# Initialize Cameras
cams = [CamThread(urls[i], f"CAM {i+1}") for i in range(4)]

prev_state = None

while True:
    frames = []
    person_detected_any = False

    for cam in cams:
        ret, frame = cam.read()

        if not ret:
            frame = np.zeros((height, width, 3), dtype=np.uint8)
            cv2.putText(frame, f"{cam.name} Offline", (40, 120),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 0, 255), 2)
        else:

            # YOLOv8 Detection

```

```

results = model(frame, verbose=False)

for r in results:
    for box in r.bboxes:
        cls_id = int(box.cls[0])
        conf = float(box.conf[0])

        if cls_id == 0 and conf > 0.5:
            person_detected_any = True

            x1, y1, x2, y2 = map(int, box.xyxy[0])

            cv2.rectangle(frame, (x1, y1), (x2, y2),
                          (0, 255, 0), 2)
            cv2.putText(frame, f"Person {conf:.2f}",
                        (x1, y1 - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.6,
                        (0, 255, 0), 2)

            cv2.putText(frame, cam.name, (10, 30),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 255, 0), 2)

    frames.append(frame)

# Send ONLY when state changes
if prev_state is None:
    prev_state = person_detected_any

if person_detected_any != prev_state:
    if person_detected_any:
        print("📡 Sending 1 (Person Detected)")
        iot.send("3", "1")
    else:
        print("📡 Sending 0 (No Person)")
        iot.send("3", "0")

prev_state = person_detected_any

```

```

# Combine 2x2 grid
top = np.hstack((frames[0], frames[1]))
bottom = np.hstack((frames[2], frames[3]))
combined = np.vstack((top, bottom))

cv2.imshow("YOLOv8 Multi-Cam (Cloud)", combined)

if cv2.waitKey(1) & 0xFF == ord('q'):
    break

# Cleanup
for cam in cams:
    cam.stop()

cv2.destroyAllWindows()
#include <KiwisIoT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define MAP_CH "1"
// ===== WIFI =====
const char* ssid = "error";
const char* pass = "12345678";
const char* topic = "dash_1774438910648";

float lat = 10.929296339763807;
float lng = 78.7381934818362;

KiwisIoT kiwisiot(ssid, pass, topic);

// ===== MOTOR PINS =====
#define IN1 D3
#define IN2 D4
#define IN3 D5
#define IN4 D6

```

```

#define LIGHT_PIN D1
#define HORN_PIN D2

// ===== SENSORS =====
#define TRIG_PIN D7
#define ECHO_PIN D8
#define IR_PIN D0

// ===== LCD =====
LiquidCrystal_I2C lcd(0x27, 16, 2);

int powerState = 0;

// Sensor variables
long duration;
int distance;
int irValue;

// SMS control
bool smsSent = false;

// ----- GSM FUNCTION -----
void sendSMS(String msg) {
  Serial.println("AT");
  delay(1000);

  Serial.println("AT+CMGF=1");
  delay(1000);

  Serial.println("AT+CMGS=\"+916374593717\""); //  PUT YOUR
NUMBER
  delay(1000);

  Serial.print(msg);
  delay(500);

```

```

Serial.write(26); // CTRL+Z
delay(5000);
}

// ----- MOTOR CONTROL -----

void motorForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void motorReverse() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void motorRight() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void motorLeft() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void motorStop() {
  digitalWrite(IN1, LOW);

```



```

digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);
}

// ----- SENSOR FUNCTION -----

void readSensors() {

  // Ultrasonic
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH);
  distance = duration * 0.034 / 2;

  // IR
  irValue = digitalRead(IR_PIN);

  // Serial output
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm | IR: ");
  Serial.print(irValue == 0 ? "ON" : "OFF");
  Serial.print(" | Status: ");

  // ----- CONDITION -----
  if (distance < 10 || irValue == 0) {

    motorStop();
    Serial.println("CAR STOPPED");

    // 🚨 SEND SMS ONLY ONCE

```

```

    if (!smsSent) {
        String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
        sendSMS("Obstacle Detected!\nLocation: " + link);
        smsSent = true;
    }

    lcd.setCursor(0, 0);
    lcd.print("CAR STOPPED ");

} else {

    Serial.println("RUNNING");
    smsSent = false;

    lcd.setCursor(0, 0);
    lcd.print("CAR RUNNING ");
}

// ----- LCD Line 2 -----
lcd.setCursor(0, 1);
lcd.print("D:");
lcd.print(distance);
lcd.print(" IR:");
lcd.print(irValue == 0 ? "ON " : "OFF");
}

// ----- IOT CONTROL -----

void onControl(String ch, String val) {

    Serial.print("CH: ");
    Serial.print(ch);
    Serial.print(" | VAL: ");
    Serial.println(val);

```

```

// ● PRIORITY: CHANNEL 3 EMERGENCY STOP
if (ch == "3") {
  if (val == "1") {
    motorStop();
    powerState = 0; // optional: also turn off power
    Serial.println("  EMERGENCY STOP (CH3)");
    String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
    sendSMS("Manual Request\nLocation: " + link);

    lcd.setCursor(0, 0);
    lcd.print("EMERGENCY STOP ");

    return; // STOP everything else
  }
}

// ----- NORMAL CONTROL -----
if (ch != "0") return;

if (val == "power:1") {
  powerState = 1;
  Serial.println("Power ON");
  return;
}

if (val == "power:0") {
  powerState = 0;
  motorStop();
  Serial.println("Power OFF");
  return;
}

if (val == "lights:1") {
  digitalWrite(LIGHT_PIN, HIGH);
  return;
}

```

```

if (val == "lights:0") {
    digitalWrite(LIGHT_PIN, LOW);
    return;
}

if (val == "horn") {
    digitalWrite(HORN_PIN, HIGH);
    delay(300);
    digitalWrite(HORN_PIN, LOW);
    return;
}

if (powerState == 0) return;

if (val == "forward") motorForward();
else if (val == "back") motorReverse();
else if (val == "right") motorRight();
else if (val == "left") motorLeft();
else if (val == "stop") motorStop();
}

// ----- SETUP -----

void setup() {

    Serial.begin(9600); // GSM uses Serial

    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);

    pinMode(LIGHT_PIN, OUTPUT);
    pinMode(HORN_PIN, OUTPUT);

```

```

pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
pinMode(IR_PIN, INPUT);

motorStop();

// LCD
lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("System Ready");
delay(1000);
lcd.clear();

kiwisiot.begin();
kiwisiot.onReceive(onControl);
}

// ----- LOOP -----

void loop() {
    kiwisiot.run();

    //  Serial command to send location
    if (Serial.available()) {
        char c = Serial.read();

        if (c == '1') {
            String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
            sendSMS("Manual Request\nLocation: " + link);
        }
    }

    static unsigned long lastSend = 0;

```

```

if (millis() - lastSend > 3000) {
    lastSend = millis();

    Serial.print("LAT: ");
    Serial.println(lat, 6);
    Serial.print("LNG: ");
    Serial.println(lng, 6);

    // Convert to string format "lat,lng"
    String loc = String(lat, 6) + "," + String(lng, 6);

    // Send to map channel
    kiwisiot.send(MAP_CH, loc);
}
readSensors();
delay(500);
}

```